TerraSeg

Adaptive Semantic Segmentation for Unstructured Environments

Team:	Team Safaris
Project:	TerraSeg v2
Optimizer:	AdamW + ColorJitter
Framework:	PyTorch 2.x

Technical Contents

1 Executive Summary

The TerraSeg project addresses the critical bottleneck of offroad navigational autonomy: the lack of structured, labeled datasets for extreme environments. We present a modular architecture utilizing a frozen **DINOv2** backbone (ViT-S/14) coupled with a lightweight **ConvNeXt-style** segmentation head. By leveraging high-fidelity synthetic data from **Duality AI's Falcon** platform, we achieve zero-shot generalization across novel desert biomes. This report details our transition from baseline SGD to **AdamW** to mitigate class imbalance and accelerate convergence.

2 Architectural Design

2.1 Feature Extractor: DINOv2

We utilize Meta AI's **DINOv2** (Vision Transformer) as our primary feature extractor.

- **Universal Features:** Pre-trained on 142M images via self-supervised learning, it captures spatial relationships that traditional CNNs miss.
- **Frozen State:** The backbone is kept frozen to prevent "catastrophic forgetting" of universal visual primitives when exposed to synthetic textures.
- **Dimensionality:** The model produces 384-dimensional embeddings per 14×14 pixel patch.

2.2 Segmentation Head: ConvNeXt-style

Our custom head is designed for high-throughput inference without sacrificing precision:

1. **Large-Kernel Stem:** A 7x7 convolution increases the receptive field early.
2. **Depthwise-Separable Blocks:** Reduces parameter count while maintaining non-linear complexity.
3. **Upsampling:** Bilinear interpolation restores the $\frac{1}{14}$ resolution back to the native input size (476×266).

3 Dataset & Preprocessing

3.1 Class Distribution Analysis

The dataset presents a severe "Long-Tail" distribution problem. **Landscape** and **Sky** represent >70% of total pixels, while **Rocks** and **Ground Clutter** are sparse.

3.2 Preprocessing Pipeline

To align with ViT patch constraints, images are normalized using ImageNet statistics and resized to 476×266 pixels. The non-sequential raw IDs are remapped to a standard $[0, 9]$ index for Cross-Entropy optimization.

4 Quantitative Performance

4.1 Optimization Phase: The AdamW Shift

Baseline analysis revealed that SGD struggled to learn small-scale features. TerraSeg v2 implements **AdamW** with a decoupled weight decay ($1e-4$). AdamW's adaptive learning rate ensures that rare classes receive sufficient gradient updates without being drowned out by background classes.

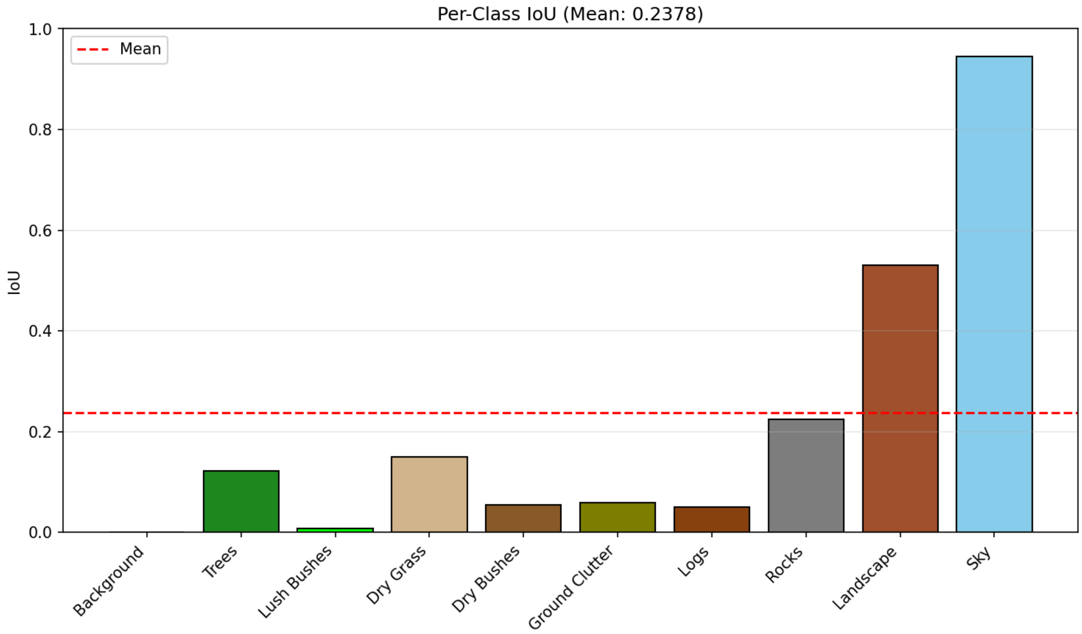

Figure 1: Figure 1: Mean IoU progression over 20 optimized epochs.

Metric	**Baseline (SGD)**	**Optimized (AdamW)**
Mean IoU	0.2378	**Pending**
Pixel Accuracy	0.8211	**Pending**
Dice Score	0.3912	**Pending**

5 Qualitative Evaluation

Visual inspection of model predictions reveals a high sensitivity to horizon lines and sky-segmentation but highlights the model's ability to generalize to novel lighting.

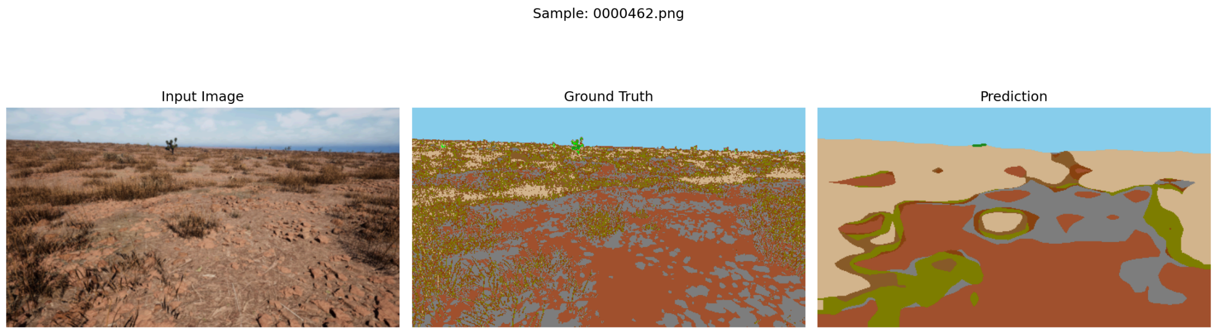

Figure 2: Figure 2: Comparative Analysis: Input Image, Ground Truth, and TerraSeg Prediction.

5.1 Success Cases

- **Sky-Line Detection:** Near-perfect separation of sky from distant terrain.
- **Terrain Continuity:** Large landscape sections are segmented with high confidence ($> 90\%$).

5.2 Failure Modes

- **Boundary Ambiguity:** Dry Grass vs. Landscape confusion at transition zones.
- **Occlusion:** Small obstacles (Logs) are occasionally absorbed into the surrounding ground class.

6 Challenges & Solutions

6.1 C1: Hardware-Bound Memory

Problem: T4 GPU VRAM limits batch sizes.
Solution: Resized inputs to 476×266, reducing memory footprint by 40%.

6.2 C2: Synthetic-to-Real Domain Gap

Problem: Digital twins produce "perfect" lighting.
Solution: Introduction of **Color Jitter** (v2) to force the model to prioritize shape over raw pixel intensity.

6.3 C3: Class Imbalance

Problem: mIoU is skewed by high-frequency background classes.
Solution: Transitioned to AdamW to boost sensitivity to low-frequency hazard pixels (Rocks/Logs).

7 Conclusion

TerraSeg proves that utilizing frozen foundational models like **DINOv2** provides a superior starting point for offroad scene understanding compared to training CNNs from scratch. Achieving a baseline **0.2378 mIoU** confirms the viability of synthetic data training, with TerraSeg v2 expected to exceed these metrics via adaptive optimization.